$i=1$

По критерию χ^2: $\Delta \rightsquigarrow \chi^2(10-2-1) = \chi^2(7)$

p-value $= P(\Delta \geq \tilde{\Delta} \mid H_0) = \int_{16,87}^{+\infty} q(t)dt = 0,018 < \alpha$

$\Rightarrow$ отвергаем H_0

[11] $H_0: \xi \sim p_0(x) = 1\{(0,1)\}$

$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1}\{(0,1)\}$

а) $n=1$, α: $l(x) = \frac{L_1(x)}{L_0(x)} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C$

$\Rightarrow G: x \leq A$

$P(x \leq A \mid H_0) = \alpha = \int_0^A dx \Rightarrow A = \alpha \Rightarrow G: x \leq \alpha$

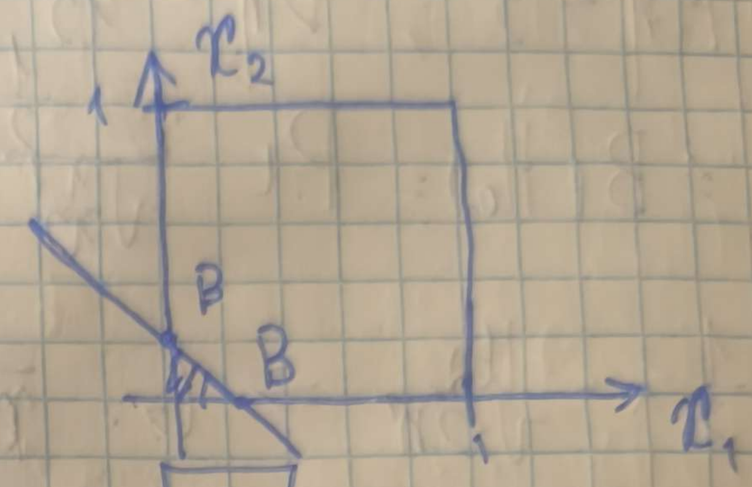
$W = P(x \leq A \mid H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1}(1-e^{-\alpha})$

$\alpha_2 = 1 - W = \frac{e}{e-1}(e^{-\alpha} - 1) + 1$, $\alpha_1 = \alpha$

b) $n=2$: $L = \frac{L_1}{L_2} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$

$G: x_1 + x_2 \leq B$

$P(x_1 + x_2 \leq B \mid H_0) = \alpha \Rightarrow \iint\limits_{\substack{x_1+x_2 \leq B \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1}} dx = \frac{B^2}{2}$



$\Rightarrow \frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha} \Rightarrow G: x_1 + x_2 \leq \sqrt{2\alpha}$

$W = P(x_1 + x_2 \leq B \mid H_{a_1}) = \iint\limits_{\substack{x_1+x_2 \leq B \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1}} \frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1} dx_1 dx_2 =$

$= \left(\frac{e}{e-1}\right)^2 \int_0^B e^{-x_1} dx_1 \int_0^{B-x_1} e^{-x_2} dx_2 = \left(\frac{e}{e-1}\right)^2 \int_0^B e^{-x_1}(1 - e^{-B+x_1}) dx_1 =$

$= -\frac{Be^2 e^{-B}}{(e-1)^2} + \left(\frac{e}{e-1}\right)^2 (1 - e^{-B}) = \left(\frac{e}{e-1}\right)^2 [1 - e^{-B} - Be^{-B}]$

$\alpha_1 = \alpha_2 \quad \alpha_2 = 1 - W$

c) асимп. критерий по выборке объема n

$L(\vec{x}_n) = \frac{L_1}{L_0} = \prod_{i=1}^{n} \frac{p_1(x_i)}{p_0(x_i)} \geq C$

$\ln L = \sum_{i=1}^{n} \underbrace{\ln \frac{p_1(x_i)}{p_0(x_i)}}_{\eta_i} \geq \ln C$

$\sum_{i=1}^{n} \ln p_1(x_i) - \sum_{i=1}^{n} \ln p_0(x_i) = \sum_{i=1}^{n} \ln\left(\frac{e^{1-x_i}}{e-1}\right) - 0 \geq \ln C$

$\sum_{i=1}^{n} (1 - x_i) - n \ln(e-1) \geq \ln C$

$G: \sum_{i=1}^{n} x_i \leq A \Rightarrow G: \bar{x} \leq B$

$P(\bar{x} \le B \mid H_0) = \alpha$

ЦПТ: $\frac{\bar{x} - M\eta}{\sqrt{D\eta}}\sqrt{n} \sim N(0,1)$

$H_0: M\eta = \frac{1}{2}$; $D\eta = \frac{1}{12}$

$P(\bar{x} \le B \mid H_0) = P\left(\frac{\bar{x} - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n} \le \underbrace{\frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n}}_{\le u_\alpha}\right)$

$B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} \Rightarrow G: \bar{x} \le \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$

$W = P(\bar{x} \le B \mid H_1)$; $H_1: M\eta = \int_0^1 x p_1(x)\,dx = \frac{e-2}{e-1}$

$M\eta^2 = \int_0^1 x^2 p_1(x)\,dx = \frac{2e-5}{e-1}$

$D\eta = \frac{e^2 - 3e + 1}{(e-1)^2}$

$W = P(\bar{x} \le B \mid H_1) = P\left(\frac{\bar{x} - M\eta}{\sqrt{D\eta}}\sqrt{n} \le \underbrace{\frac{B - M\eta}{\sqrt{D\eta}}\sqrt{n}}_{=\gamma}\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$

где $\gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - \frac{e-2}{e-1}}{\sqrt{\frac{e^2 - 3e + 1}{(e-1)^2}}}\sqrt{n}$

$\alpha_1 = \alpha$, $\alpha_2 = 1 - W$

$\gamma \xrightarrow{n\to\infty} +\infty \Rightarrow W_n \xrightarrow{n\to\infty} 1$ $\forall$ распр. из $H_1 \Rightarrow$ критерий состоятелен

d) $G: x_{min} < C$

$P(\vec{x}_n \in G \mid H_0) = \alpha$; $\xi_{min} \sim 1 - (1 - F_0(x))^n$, где $F_0(x) = x \in [0,1]$

$P(x_{min} \le C) = 1 - (1 - F(C))^n = \alpha$

$\alpha - 1 = -(1 - C)^n \Rightarrow C = 1 - \sqrt[n]{1-\alpha}$

$G: x_{min} < 1 - \sqrt[n]{1-\alpha}$

$W = P(x_{min} \le C \mid H_1)$

$F_1(x) = \int_0^x \frac{e}{e-1} e^{-t}\, dt = \frac{e}{e-1}(1-e^{-x})$

$\Rightarrow W = P(x_{min} \le C \mid H_1) = 1 - (1-F_1(C))^n =$

$= 1 - (1 - \frac{e}{e-1}(1 - e^{-1+\sqrt[n]{1-\alpha}}))^n =$

$= 1 - (1 - \frac{e}{e-1} + \frac{e^{\sqrt[n]{1-\alpha}}}{e-1})^n$

$e^{\sqrt[n]{1-\alpha}} = \exp(e^{\frac{1}{n}\ln(1-\alpha)}) = \exp(1 + \frac{1}{n}\ln(1-\alpha) + o(\frac{1}{n})) =$

$= e(1 + \frac{1}{n}\ln(1-\alpha) + o(\frac{1}{n})) \Rightarrow$

$\Rightarrow W = 1 - (1 + \frac{e + \frac{e}{n}\ln(1-\alpha) + o(\frac{1}{n})}{e-1} - \frac{e}{e-1})^n =$

$= 1 - (1 + \frac{e\ln(1-\alpha)}{e-1} \cdot \frac{1}{n} + o(\frac{1}{n}))^n \xrightarrow{n\to\infty} \exp(\frac{e}{e-1}\ln(1-\alpha)) =$

$= 1 - (1-\alpha)^{\frac{e}{e-1}} \ne 1 \Rightarrow W \underset{n\to\infty}{\not\to} 1 \Rightarrow$ критерий не явл. состоятельным

[12] $H_0: p_1 = \frac{1}{3},\ p_2 = \frac{1}{6},\ p_3 = \frac{1}{4},\ p_4 = \frac{1}{4}$

$H_1: p_i = \frac{1}{4}$

$n = 2,\ \alpha = 0{,}2$

$L = \frac{L_1}{L_0} = \frac{(\frac{1}{4})^2}{p_{1i}p_{2i}} = \frac{1}{16\, p_{1i}p_{2i}} = \frac{(\frac{1}{4})^2}{p_{0i}p_{0j}} = \frac{1}{16\, p_{0i}p_{0j}}$

Чем больше L, тем больше свидетельств в пользу H_{01}

Исходы	p_{0i}, p_{0j}	l
(3, 3)	$\frac{1}{36}$	2,25
(1, 3), (2, 3), (3, 1), (3, 2)	$\frac{1}{24}$	1,5
(3, 4), (4, 3)	$\frac{1}{18}$	1,125
(1, 1), (1, 2), (2, 1), (2, 2)	$\frac{1}{16}$	1
(1, 4), (2, 4), (4, 1), (4, 2)	$\frac{1}{12}$	0,75
(4, 4)	$\frac{1}{9}$	0,5625

Упорядочим l по возрастанию l и накопим вероятности при H_0

- $l = 2{,}25$: (3,3), $p_0 = \frac{1}{36} \approx 0{,}0278$
- $l = 1{,}5$: (1,3), (2,3), (3,1), (3,2), $p_0 = 4 \cdot \frac{1}{24} \approx 0{,}1667$

$P = \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \approx 0{,}1944 \le 0{,}2$

При добавлении следующей группы $P = \frac{7}{36} + \frac{1}{9} \approx 0{,}3056 > 0{,}2$

$\Rightarrow G = \{(3,3); (1,3); (2,3); (3,1); (3,2)\}$ – критическая область

$\alpha_1 = P_0(G) = \frac{7}{36}$ – ошибка первого рода

$W = (\vec{x}_n \in G \mid H_1) = 5 \cdot \frac{1}{16} = \frac{5}{16}$ – мощность критерия

[13] $\xi \sim N(a, \sigma_\xi^2)$ $\quad n = 139$

$\eta \sim N(b, \sigma_\eta^2)$ $\quad m = 1000$

Длина: $S_\xi^2 = 5{,}722$; $S_\eta^2 = 6{,}161$

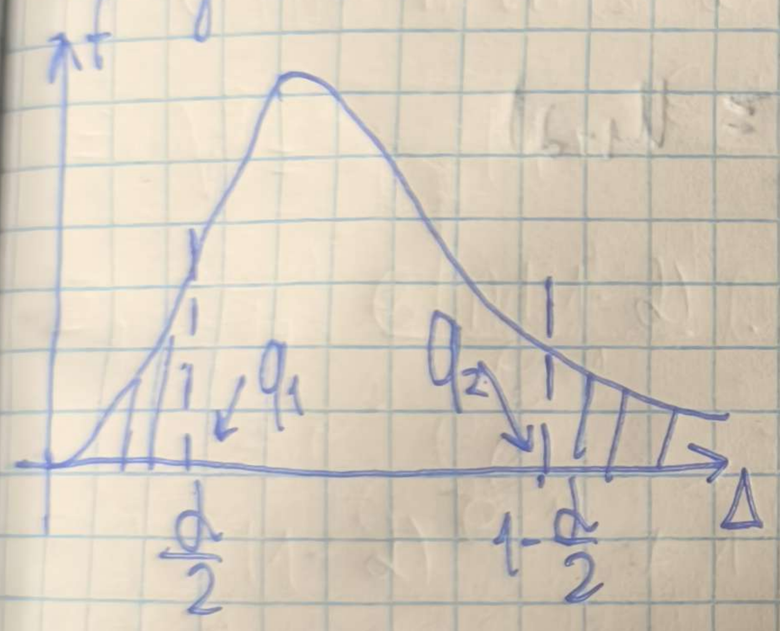
Ширина: $S_\xi^2 = 4{,}612$, $S_\eta^2 = 5{,}055$

$H_0: \sigma_\xi = \sigma_\eta$, т.е. $\frac{S_\xi^2}{S_\eta^2} \sim F(n-1, m-1)$

$H_1: \frac{\sigma_\xi}{\sigma_\eta} = \nu \neq 1$, т.е. $\frac{S_\xi^2}{\nu^2 \cdot S_\eta^2} \sim F(n-1, m-1)$

Тогда

$$\Delta = \left(\frac{S_\xi}{S_\eta}\right)^2$$



$$G: \left[\begin{array}{l} \Delta \geq q_2 \\ \Delta \leq q_1 \end{array}\right.$$

$$\Rightarrow P\left(\left[\begin{array}{l} \Delta \geq q_2 \\ \Delta \leq q_1 \end{array}\right. \middle| H_0\right) = \alpha$$

$\Delta \sim \alpha^2 F(n-1, m-1)$ – функция распр. Фишера

$$W = P\left(\left[\begin{array}{l} \Delta \geq q_2 \\ \Delta \leq q_1 \end{array}\right. \middle| H_{01}\right) = \tilde{f}\left(\frac{q_1}{\alpha^2}\right) - 1 - \tilde{f}\left(\frac{q_2}{\alpha_0^2}\right)$$

$\tilde{f}$ – плотность распр. Фишера

2 [14] $x \sim N(a, \sigma_x^2)$, $\sigma_x^2 = 2$

$y \sim N(b, \sigma_y^2)$, $\sigma_y^2 = 1$

1:

a_1: $H_0: a = b$; $H_1: a = b$

13

1:

$\Delta = \bar{x} - \bar{y} - (a - b) \sim N(0, \frac{2}{n} + \frac{1}{m})$

1:

1 Если H_0 верна, то $\Delta = \bar{x} - \bar{y} \sim N(0, \frac{2}{n} + \frac{1}{m})$

1:

1 В H_1 интересует только $a > b \Rightarrow G = \{\Delta \geq u_{1-\alpha}\}$

1

1 $]\ a - b = \gamma \Rightarrow \tilde{\Delta} = \bar{x} - \bar{y} - \gamma \sim N(0, \frac{2}{n} + \frac{1}{m})$

1

1 $\bar{x} - \bar{y} \sim N(\gamma, \frac{2}{n} + \frac{1}{m}) \Rightarrow W(\gamma) = P(\hat{\Delta} \geq u_{1-\alpha})$